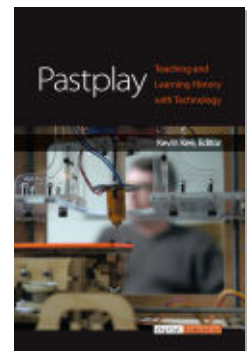




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13. Teaching History in an Age of Pervasive Computing: The Case for Games in the High School and Undergraduate Classroom



Published by

Kee, Kevin.

Pastplay: Teaching and Learning History with Technology.

University of Michigan Press, 2014.

Project MUSE. <https://doi.org/10.1353/book.29517>.

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[193.151.14.33] Project MUSE (2024-07-31 15:16 GMT)

Teaching History in an Age of Pervasive Computing

The Case for Games in the High School and Undergraduate Classroom

Kevin Kee and Shawn Graham

Historians have always been interactive with the content that we study, constantly challenging, reworking, and indeed, remixing information to “do history.” And we have incorporated that interactivity into our teaching, analyzing primary and secondary sources with our students in seminars, and helping students draw on those sources to craft their own historical narratives. The arrival of computer technologies has provided new ways to support interactivity in our teaching.

Our students require it: there has always been a world wide web for the undergraduates in our classes. Personal computers were first introduced en masse into primary and secondary education in the 1980s, and those students have already graduated from university. Computers went, in the span of a few years, from being a rarity to a commonplace. We now live in an age of “pervasive computing,”¹ in which digital devices proliferate into every corner of our lives. Students interact with this technology less like a tool (something to get the job done) and more like a musical instrument (something with which to be creative). The key aesthetic of computing today is not key-boarding, or re-creating previous media in digital format, but rather, content creation, mash-ups, and remixes: in short, interactivity.² Several years ago a 2007 Demos Report surveyed primary- and secondary-level students and parents in the United Kingdom and created focus groups to study the digital impact of new media on their day-to-day lives and especially their learning environments. The key finding was that for young people (today’s university

students) the new media tools were used to strengthen existing social networks and to create expressive content.³

How do we teach history in an age of pervasive computing, where interactivity with (rather than consumption of) media in the context of social networks (rather than in isolation) is key? Not through “websites” or “bulletin board forum posts.” These are interim technologies—what the historian John Sutton Lutz called the “horseless carriages” of the computer revolution. Instead, we need to progress to “the automobile.” One phrase expressed the new invention in terms of existing technology; the other coined a completely new idea to describe the technology. Just as the arrival of “the automobile” coincided with mass production and mass access, the new way of interacting with digital media has started to create its own idioms and metaphors. Social apps. Facebooks and Machinima. MMORPGs (massively multiplayer online role-playing games).

These last two terms are connected to computer games, the most exciting, technically demanding, computing applications today. They are the digital media “automobiles” of the twenty-first century. Game technologies have driven the development and evolution of computer hardware, artificial intelligence, database management, and a host of allied technologies. Computer games are some of the most complicated and sophisticated simulations available, with design and development budgets that dwarf those of many movies, and certainly any Humanities Department’s research budget. As a result, game studies are growing, but the nascent discipline is dominated by computer science and psychology research; the humanities have had relatively little to offer.

While the humanities have shown limited interest in games, games have shown great interest in the humanities, and especially in history. A recent survey showed that 26 of the 133 PC-based games that have sold at least 1 million units have been based on a historic theme, or have employed historical tropes.⁴ Clearly, given that a fifth of the all-time best-selling computer games have historical themes, there is room for humanities- and history-based analyses of computer games, and consideration of how best to use this popularity to further the teaching and learning of history.

We intend to go further. We believe that the best way to teach history in an age of pervasive computing is through collaborative learning with computer games. This chapter is divided into four parts. We begin by suggesting that games should be used in our undergraduate courses in much the same way that we have used texts. History games are synthetic historical worlds, similar to the narratives on our class reading lists, except that these are expressed in computer code, not language. While the academic literature has championed games as a teaching “tool,” we take a different view: that

these are artifacts that should be deconstructed, in the manner of historiography. But how do we know which history games to use? The marketplace has made claims for history games that must be challenged, and we propose a specific typology by which to understand the place of history games in our undergraduate courses.

In the second part of the chapter, we show how students can build on their analysis of games by creating their own histories through game “mods” (modifications of commercial games). The process is similar to that which sees students build on their analysis of texts to write historiographical essays and benefit from peer review. Examples from web forums, and our own experience, highlight the potential for peer review. In the third part of the chapter, we draw on our own experience to show how students can move beyond analysis, and modding, to collaboratively developing their own games, in much the same way that they write research papers. Finally, we reflect on our use of games for history to suggest how we might best assess the work of our students. In these ways, we show how historians can tap the potential, while avoiding the pitfalls, of learning with games.

Narratives and Games as Synthetic Worlds

A conventional history course requires that a student engage in the literature related to the topic. In both lecture and seminar courses, students read in preparation for small-group discussions, guided by an instructor or teaching assistant. Historians who want to use technology in an age of pervasive computing can use computer games in the same way that we have previously used books and articles.

Those books and articles are worlds that we have created, drawing on evidence from the past that has been preserved in the archives. The past is disorganized, meaningless, and exists beyond the rules of language. History is organized, meaningful, and expressed with the rules of language.⁵ Created by historians writing in the present moment and therefore occupied with present concerns,⁶ and written in narrative form, our histories follow an artificially linear path, with a beginning, middle, and end. We ask our students to immerse themselves in these synthetic worlds, and draw from them insights that they can apply to their understanding of the topic at hand.

In a similar way, as the game theorist Edward Castranova has pointed out, a game is a synthetic world. But where historians’ books and articles make a persuasive case through narrative, games are compelling because these practice “just-good-enough” virtual reality. As Castranova notes: “a game perspective focuses all thought and research on the user’s subjectivity

and well-being. It insists on immediate usability. It thrives on widened access and multiple users. And it generates a willing suspension of disbelief, without which genuine immersion cannot happen.⁷ If a game is effective, it immerses a player, so that she projects her mind—her sense of self—into it. From these experiences, as the linguist and game theorist James Paul Gee has pointed out, gamers learn a great deal. Indeed, according to Gee, games are one of the most effective teaching tools yet devised.

The challenge for historians is that, with a few notable exceptions, history games are not created by researchers focused on learning; they are built by gamers obsessed with fun. But that does not make them a waste of time. Just as we ask our students to assess “popular history,” so too can we use popular games for the purposes of learning. Indeed, our task in the age of pervasive computing is to reconcile these two kinds of synthetic worlds. But how do we assess the suitability of a game for history? How do we know which games to put on our “gaming list”? Presently, the term “history game” is used to denote many different kinds of experiences with computer media. If we are going to be clear about how collaborative learning with computer games can teach history in a new era of pervasive computing, we need to clarify what we mean by “history games.” Alas, the marketplace has only muddled the issue.

History Games in the Marketplace—The Genre Problem

The type of game (the way it is played, its structure) is how the vast majority of games are classified and marketed. Games are usually discussed via a comparison of one game to another or by reference to a genre.⁸ Genre in games usually refers to the gameplay mechanics from the point of view of the player, such as first-person shooter or role-playing game. These categories are not overly useful for understanding how historically themed games could be employed by a professor or student since many of these kinds of categories are artifacts of the technology used to deliver the game. First-person shooters evolved from video arcade games to home consoles such as the Sony Playstation; role-playing games evolved from text adventure games to home PCs. As these technologies have developed, taking on qualities of one another, the genre categories have begun to overlap as well. Most of today’s first-person shooters, for example, contain many of the attributes of role-playing games.

In a marketplace saturated with thousands of first-person shooters and role-playing games, game publishers have attempted to distinguish their products by covering the gameplay mechanics with a façade of content. As a

result, we have first-person shooters set far in the future, such as the highly popular *Halo* series, and others based in the past, such as the equally successful *Call of Duty* franchise. Ask a 15-year-old if he plays history games, and he may catalogue the German soldiers he killed in his attempt to defend Chambois, ca. 1944. There is history learning here, but it is incidental. And any historian who attempts to use *Call of Duty* to teach history will quickly realize the limitations of the product. At the end of an hour, this history game is essentially about shooting people.

The frustration with history games that results from genre confusion is evident in discussions surrounding another popular franchise, and one that is marketed as specifically historical: *Civilization*. The first wave of research into history games for learning pointed to the potential of *Civilization* as a tool in the classroom.⁹ Kurt Squire has expanded this focus, with an emphasis on its effectiveness with elementary students in concert with other tools, such as encyclopedias.¹⁰ But other researchers have criticized the game's implicit narrative of technological progress as the prime mover of history, and questioned its appropriateness for history education.¹¹

Civilization and other comparable games are, according to the conventional genres, turn-based strategy games, where each turn builds on the actions taken in previous turns. In the case of *Civilization*, the player guides a tribe of people from the Stone Age to the Space Age, conquering the world as she goes. It would be hard to imagine an alternative conception of historical process being built into a turn-based strategy game—the mechanics of the game are built for “progress.” Historical contingency has been determined by the formal rule system, which has been created by the computer programming.¹²

The point, as other critics have noted, is that the game, or any computer game for that matter, is ultimately about mechanics, and not about content. The content is window dressing, and deep playing of a game such as *Civilization* teaches little about history, but everything about how to manipulate the complex algorithms that model the simulation. As Robert MacDougall points out,

Civilization's game play erases its own historical content. Learning to play means learning to ignore all the stuff that makes it a game about history and not about, say, fighting aliens. One could easily program a different game with a different set of ideological assumptions—Galloway imagines a “People's *Civilization*” game by Howard Zinn—and see precisely the same de-historicizing effect. Mastering the simulation game necessarily involves a journey away from reality towards abstraction, away from history towards code.¹³

History Games for Historians—A Typology of Time and Space

If a so-called history game primarily teaches a player how to win, why bother? The answer is that there is much to learn about history through an understanding of the history game's programming. As a result, those of us who are interested in using games for history learning need to focus on the computer code, rather than on the marketing hype or content façade. The code determines the rules of the game (the way it operates). And if the rules promote a particular way of looking at the world—if they make an argument in code for a particular worldview (what Ian Bogost calls “procedural rhetoric”¹⁴)—then we need to understand which rules, which games, best embody the historical epistemologies we wish to teach. We also need to imagine the possibilities beyond *Civilization*, including modified games, or new games, which could manifest the epistemologies we want to express. Following William Urrichio, we need to “think of the rule systems that characterize various brands of history as constituting the potential rule systems for game play.” Addressing the criticisms that have been leveled at *Civilization*, he points out that “by embedding various historiographic epistemologies as structuring agencies rather than relying implicitly on *Civilization*'s narratives of truth, progress, and the American way, a new dimension could be added to play, more coherently addressing history.”¹⁵

The first step to “coherently addressing history” in play is to determine which epistemology to teach. The possibilities are numerous, the focus of a rich vein of literature, and outside the focus of this chapter. Presuming that the historian knows what it is that he wants to teach, we can move to the second step: understanding the power of different games for addressing or reinforcing different kinds of history. To accomplish this, we need to replace the marketing hype and content façades with a clear and unambiguous typology. One such typology organizes games according to their relationships to goals.¹⁶

On one side are games as goal-oriented challenges; the mental challenge provides the fun. Too easy, and the game is boring; too hard, and the game is frustrating; to be in the sweet spot between the two is to be in a state of “flow.”¹⁷ On the other side are goalless games such as *The Sims*, or heavy management simulations like *RailRoads!* For Jesper Juul, such “goalless” games, or games without set end-states, allow the player to push social norms (deviant behavior in *The Sims*) or express personal aesthetics (like making the most beautiful city in *Caesar IV*).¹⁸ Goal and goalless is still too broad and nebulous a foundation on which to build a typology because a single game might have aspects of both, and an effective typology must create unambiguous categories. Alternatively, Espen Aarseth et al. have developed a typology

of games (not just computer video games) that considers games according to spatial movement.¹⁹ Broadly, this open-ended typology depends on classification of movement along five axes: space, time, player-structure, control, and rules. Each of those categories can be further broken down, but for our purposes we will concentrate only on the categories of space and time, as per table 13.1. By considering movement as the basis for the typology, Aarseth and his colleagues eliminate the possibility of overlaps, which is what a good typology must do. They also focus attention on how the game treats time and space. Historians are trained to move through time, if only in our minds. The narrative “synthetic worlds” that we produce out of our travels—books and articles—reflect those journeys. Historians who use games, these “synthetic worlds” supported by computers, which immerse the player in a “good enough reality” that mimics 3D space, must necessarily move through time and space as well.

How can this typology help us determine which games we might “assign” to our students to play, and later analyze in class? Let us compare two games that were released within months of each other, and that we have used with students: *Civilization IV* (published in November 2005) and *Caesar IV* (a rival game that was launched soon after). Both of these games take us into a “synthetic world” loosely based on antiquity. *Gamerankings*, a highly popular website that ranks computer games and classifies them according to “genre,” treats these as essentially the same: *Caesar IV* is listed as “Strategy, City Building, Historic,” while *Civilization IV* is categorized as “Strategy, Turn-Based, Historic.” There is nothing mutually exclusive about these categories: a “City Builder” game could be “Turn-Based,” and *Civilization IV* can be played in a concurrent, non-turn-based mode, in its multiplayer version. The genre classifications make these games appear similar, and tell us nothing about their underlying epistemologies or procedural rhetorics.

When we break them down according to Aarseth et al.’s movement typology, the differences become clear (table 13.2). *Caesar IV* is a city-management simulation based in ancient Rome. In terms of space, the view is “Omni-present”; no part of the game environment is unknown to the player. In

TABLE 13.1. Aarseth et al.’s game typology

Space	Perspective	Omni-Present / Vagrant
	Topography	Geometrical / Topological
	Environment	Dynamic / Static
Time	Pace	Real-Time / Turn-Based
	Representation	Mimetic / Arbitrary
	Teleology	Finite / Infinite

TABLE 13.2. *Caesar IV* and *Civilization IV* according to Aarseth et al.'s game typology

		<i>Caesar IV</i>	<i>Civilization IV</i>
Space	Perspective	Omni-Present	Vagrant
	Topography	Topological	Geometrical
	Environment	Dynamic	Dynamic
Time	Pace	Real-Time	Turn-Based
	Representation	Arbitrary	Mimetic
	Teleology	Finite	Finite

Civilization IV, in contrast, much of the early gameplay is built on exploration and discovery, and actions by other “civilizations” that occur off-screen can affect the player; the “Perspective” is “Vagrant.” In *Caesar IV*, the player interacts with the “Topography” by placing buildings or other structures in limited areas (“olive groves,” for instance, can only be placed on “farm land”), and so the topography is “Topological,” whereas in *Civilization IV* the player may move the game pieces almost anywhere, and so the topography is “Geometrical.” The “Environment” of both games is dynamic; it changes according to, but sometimes regardless of, the player’s actions.

Consider also what Aarseth and his colleagues call the “Pace” of time within both games: in *Caesar IV*, time moves forward regardless of the player’s actions, so the game is played in “Real-Time.” In *Civilization IV* time stops while the player moves his pieces around the board, so its time is “Turn-Based” (each player must wait for the other players to complete their turn, as in *Monopoly*). The “representation” of time, as Aarseth et al. frame it, differs in both games as well. In *Caesar IV*, if a player has the *denarii*, a brand-new Coliseum can instantly be placed within a city (time is “Arbitrary”), whereas in *Civilization IV* time is “Mimetic” (imitated), so it takes a number of turns, reflecting something of the actual cost in time, to build a Coliseum. Finally, addressing what Aarseth and his colleagues call the “Teleology” of time, *Caesar IV* is a finite game: it has a definite end point (when mission goals are reached). *Civilization IV* is also a finite game: famously, it ends when you launch a colonization mission to the stars (although other end-games are possible, including the annihilation by, or of, your foes).

An analysis of these axes can help us better understand when and how to use these games with our students. In the case of *Caesar IV* and *Civilization IV*, the different treatments of perspective and the representation of time suggest that *Caesar IV* would be helpful in addressing specific issues: exploring microeconomics of cities or the role of religious belief in urban life; rebuilding specific real-world cities, to contrast what the game suggests about how life was lived in the past versus current historical thinking, or the

current understanding of the archaeological record. *Civilization IV*, on the other hand, would be better suited for exploration of large-scale issues: diffusionism as a theory in cultural evolution (and the historiography of diffusionism), the dynamics of Roman civil wars, or the emergence of city-states in different climatic conditions.

This kind of analysis, using a typology of movement, can also help students understand why and how a game forces them to think along certain paths. The game becomes, not a teaching tool, but a kind of artifact that must be studied to determine its procedural rhetoric, which can then be deconstructed in the tradition of historiography. In the same way that we teach our students to recognize an author's viewpoint, and to analyze a text, we can teach our students to recognize implicit points of view in a game, and to analyze the rules encoded in the programming.

Modding and the Meta-Game

But we need to go further; we must use these technologies to help our students create their own representations of history. In this section, and the one that follows, we discuss two approaches to using digital games to help students create history: first, creating new content in the context of an already existing game, and second, creating an entirely new game, *ex novo*. Both of these operate at the level of what we call the “meta-game,” and it is here that we find the greatest opportunities for teaching history in an age of pervasive computing. Meta-gaming refers, in a limited sense, to game tactics that exploit bugs or features of a game in a way that was not originally intended by the game designers—like using a glitch in the game physics to scale walls that were meant to be impassable.²⁰ We employ the term in this way, but we also use it in a larger sense, to refer to an outside-looking-in awareness of the game mechanics. This is a “gaming of the game,” in which students’ engagement with history games moves beyond treating them as artifacts that must be analyzed, to modifying and even building them for themselves. The task is similar to that faced by students in a conventional history course, in which they go beyond analyzing texts to engaging them through the act of writing. In these “literature reviews” or “historiographical papers,” students articulate a thesis, using the building blocks of professional researchers. The task is both creative—students are developing their own representation of history, and synthetic—they are drawing on the literature created by historians. In a sense, they are playing with the texts.

The most common application of meta-gaming to commercial games is to tweak, adapt, modify, or otherwise alter the original game. In the early

days of computer gaming, this was often accomplished by exploiting bugs in the game's programming. Savvy game publishers soon realized that there were commercial benefits to this activity, and many now provide in-game editors to allow the players to tweak the gameplay easily. The commercial rationale is straightforward: the more that players talk about the game, and provide additional content, the greater the "buzz" and the number of copies of the game sold. Some of these modifications (mods) become so popular that they eclipse the original game. *Counterstrike*, to take the most notable example, was a mod that became more popular than its progenitor, *Half Life*.

Civilization IV, to return to our earlier example, is one such modifiable game. Previous versions of the game allowed players to customize the map and starting conditions. The most recent version lets players change the actual rules of play, and in this way contest the procedural rhetoric of the game. Only a minority of players have the requisite skills to rewrite the rules; most settle for more cosmetic changes. *Civilization IV* distinguishes between these as "mods" (rule changes) and "scenarios" (customized starting conditions). There are a number of sites that help the player achieve these customizations, with *CivFanatics* and *Apolyton* among the most popular. Indeed, *Apolyton* even operates Apolyton University website, where players can study tutorials to increase their skill in play and modding. An informal poll of 111 participants on *CivFanatics* conducted in April 2008²¹ found that 18 percent considered themselves to be "professional historians," 25 percent considered themselves to be non-historians, while the majority saw themselves as "amateurs." These were "amateurs" in the most literal sense, "amators" or "lovers" of history, debating and discussing in a manner not out of place in a university seminar.

Like undergraduate students, "Civfanatics" present their work to their peers (game mods, rather than essays), which their colleagues play, rather than read (for review).²² Feedback from the discussions that follows is used to guide further modifications and enhancements of the mod. For example, several "Civfanatics" in 2006 engaged in a meta-game of *Civilization IV* connected to a mod set during the Crusades.²³ A participant who went by the name of "Holyone" began the discussion with a post outlining the period that he had modeled, an indication of the depth of his mod (his scenario would require several hours of intense gameplay), and the downloadable file. The forum post, written with the distinctive spelling, syntax, and grammar of the Internet, read as follows: "Holyone" posted: "The Crusades (European Middle Ages Mod). . . . The scenario is set in the Middle East during the time of the Crusades. The time period is 1100–1300 AD at Marathon speed, which means 1 year per turn. Playable civs: Kingdom of Jerusalem (Baldwin I of Boulogne); Byzantine Empire (Basil II); Egypt (Saladin); Rum Sultanate (Alp Arslan); Tatar Khaganate (Timur Lenk)."

Within days, other Civfanatics had downloaded the scenario, played it, and offered their feedback and suggestions—peer review: “Drtad” posted: “Nice work Holy One, but shouldn’t you have the Armenian Kingdom of Cilicia in their, do a Wiki search as they were important during the Crusades by letting the Crusaders pass.” “Holyone” replied: “Of course they were important! I also miss the Kingdom of Georgia, another important christian kingdom, but that part of the map is just way too crowded. Perhaps on a bigger one. I saw nice Middle East maps elsewhere, so maybe I will use one of them. But if anybody knows one, link it here, pls!” The feedback was not concerned with gameplay—after all, “Holyone” had not modified the computer code—but rather with historical accuracy: “Ohcrapitsnico” posted: “Wasn’t Salah al-din the king of the abbasids not just egypt? Secondly, are you planning for this just to be a scenario or like a mod?” “Holyone” pointed out that, given the limitations of the map (the game board, with its computer code) he could not include every “civilization” that had occupied the real-world territory. Like a student writing an essay with a predetermined page limit, he had to make choices: “Holyone” posted: “Yes he was, but it’s easier to have only one civ and leader for the different time period. Even in this short(?) 200 years there was a different Egypt when the Crusaders arrived in 1097–99, another one that is Saladin’s and when the mameluks took over is yet another story.” He justified his choice of Saladin by noting that “he is a rather emblematic figure of the Crusades (along with Richard I). But if you ask his territorial rule, in the scenario Egypt starts with the Nile valley and Mesopotamia (Baghdad, Damascus) under her rule. And also Mecca as it is the Islam holy city.”

The concern with historical accuracy continued, but with an additional focus on the need to change the game rules so that the period might be properly modeled: “Drtad” posted: “Nice work on the new map Holyone. But shouldn’t Lesser Armenia and Georgia be Orthodox? They surely were not Catholic.” “Holyone” replied: “I too thought about that, but there are sooo many branches of christianity especially that part of the world. Georgia for example is the first christian country, that time there was no orthodox or catholic christianity. Christianity in Syria, even today, is neither, but (you are right) more close to orthodoxy. The patriarch of Antioch and Alexandria had great dogmatic debate with the Patriarch of Constantinople, and they practically broke up. I did choose Catholicism because of the diplomatic relation bonus, but I can still change one of them (or both), if it suits you better.” “Drtad” answered with an admonishment: “Georgia was definitely not the first Christian nation of the world. That title belongs to Armenia. 301 AD if I am not mistaken. King Drtad adopted Christianity after St. Gregory cured him of a didease just by touching his forehead.”

“Holyone” responded by creating a new map and instituting several rule changes. *Civilization* models religious influence in a specific manner (civilizations with the same religion tend to be allies) so the debate concerning Orthodox or Catholic Christianity in Georgia was significant not just for historical accuracy, but also for gameplay. The debate over content forced a debate over the computer code, and a resulting alteration, which spurred further discussion. The act of creation and exchange was not unlike that which occurs in a university seminar. Instead of writing a historiographical paper using the publications of historians, “Holyone” created a mod drawing on the game *Civilization IV*. Instead of presenting it to a class, and engaging in a debate with his colleagues, he published it to the web, where his peers could respond, with passionate arguments, expressed in misspellings, emoticons, and other signifiers of the Internet age. “Drtad” was a significant player in this meta-game, and not just because he had taken the name of the king of the first country to adopt Christianity. As a result of his comments, and those of the rest of the community, “Holyone” modified and enhanced his scenario, resulting in a more persuasive product of historical play.

This kind of peer review is not unusual. In “The Rise and Fall of Rome,” another Civfanatics discussion, amateur historians, not historians or classic students, developed a historically “authentic” simulation of Roman culture. They explored the conditions behind the emergence of the Social War (the war between Rome and its Italian allies, or *socii*, during the early first century B.C.E.), and devised a way to allow for the possibility of the war to emerge out of gameplay.

These examples from the meta-game surrounding *Civilization IV* demonstrate how we envision teaching through gaming. As we showed in the first part of the chapter, the game itself becomes not a teaching tool, but rather a kind of artifact that we study to determine its procedural rhetoric, which we then deconstruct in the tradition of historiography. Students can then build on this analysis, as we show immediately above, and through modding and the meta-game, create their own representations of history.

Building a Game ex novo, and the Meta-Game

If historians understand computer code as another language through which to express historical thinking, we can move beyond tweaking the algorithms of an established game such as *Civilization*. Taken to its logical conclusion, the ultimate meta-game is the construction of a game ex novo, in which code is used to develop an original artifact. In the same way that students write a research paper, investigating primary and secondary sources and then

assembling an essay in an effort to persuade the reader of a thesis, students can develop games, engaging the sources and then building a system to argue for a specific explanation of history.

Educators have long pointed out the benefits of building knowledge and understanding in this manner. Led by Seymour Papert, theorists have advanced the notion of “constructionism” (a term coined by Papert and Idit Harel,²⁴ promoting the construction of knowledge in the mind of the learner). In the field of computing, Papert drew on the work of Jean Piaget to develop the Logo programming language for students, so that they could write and execute basic programming functions, including the programming of robots.

In the field of history, students can use computer code as a language to express historical thinking, through games. In a 2007/08 upper-level undergraduate course project taught by Kevin Kee at Brock University in St. Catharines, Ontario, students opted to use C++, a general-purpose programming language, with Open GL (Open Graphics Library), a set of procedures enabling a computer’s operating system to produce 2D and 3D graphics. Programming in C++ is beyond the knowledge and abilities of the vast majority of humanities undergraduates, so the course project included upper-level undergraduate computing science students.

The history students split their time between conventional seminars, tutorials, and design sessions. The seminar required that they read and discuss the epistemology of history, the manner in which computers have provided new opportunities to express these epistemologies, the essential debates in the new field of games for learning, as well as project management and game design. The computing science students, for their part, split their time between traditional lectures, and joined the history students in the tutorials and design sessions. They focused on the challenges faced by computing scientists collaborating on software projects, including working as members of a team.

In groups of six to eight students, the history and computing science majors began by brainstorming a game concept, focused around the War of 1812 in Niagara. They next defined the goal of their game, and drafted a proposal that included the game description and an overview of the content. The proposal also incorporated documents relating to their work together as a team: a contract among the team members, a chart outlining the phases of their work, and a schedule of deliverables. A “Design Document” followed, specifying aspects of the game (such as its target audience and its technical requirements). Only after these steps were complete did they begin development. The final class saw them “launch” their games online, with family, friends, and colleagues in attendance.



Fig 13.1: *Brigade: A War of 1812 Saga*. Screenshot courtesy of the authors.

The results of their work paled in comparison to a *Civilization IV* mod. The graphics were simple, and the gameplay restricted. But what these lacked in complexity and depth, they made up for in originality. Some of the games, such as *Brigade: A War of 1812 Saga* (see figure 13.1 above) addressed key battles in the war, enabling players to re-fight the conflicts from the American or British side.²⁵ The results provided opportunities to examine counterfactuals, including how different battle strategies might have affected the outcome of the war. Other games addressed the economics of the war, including how merchants tried, and sometimes failed, to deal with a changing financial environment. *Tavern Keeper* (see figure 13.2 below), for instance, put the player in the apron of the proprietor of one English Canada's oldest inns, in Niagara-on-the-Lake.²⁶ Serving drinks from behind the bar brought the player into contact with battle-weary British soldiers, anxious merchants, and concerned farmers.

The students' goal, one of the developers noted afterward, was to combine political history, with its focus on geopolitical events (the War of 1812), with social history and its concern with the lives of "ordinary citizens" (the tavern keeper). The end result was a game that provided an omnipresent perspective, allowing the player to see everything, and gain confidence in this new environment. The player had relatively little freedom to move around the tavern, and none beyond it—the space was topological. The environment was static—while the player could influence the environment (cultivating a



Fig 13.2: *Tavern Keeper*. Screenshot courtesy of the authors.

relationship with a patron, for instance) the object of the changes was altered in number only; the environment itself remained essentially unchanged. In terms of time, the game was turn-based: the player made a decision (for example, to reinvest profits into the tavern, or to use them to diversify the business and purchase land), and then awaited the consequences of her actions. The representation of time was mimetic; events occurring in the game mimicked the corresponding time in the real world—cultivating a relationship with an influential patron might yield valuable, insider information later in the game. Finally, *Tavern Keeper* was teleologically finite—the player eventually reached a final win or lose state.

The omnipresent, topological, static space of the tavern limited the movement of the player, and concentrated her attention on a specific place. The turn-based, mimetic, finite timeframe rooted the player in the war period, while providing her with time to consider carefully her choices and grasp the complexity of her changing economic environment. In this way, the student leader noted, “the game showed a single individual merchant’s experience on a cultural and personal level and then put it into a large context by linking it to the War of 1812, showing why it is important. With a game we were able to show a nuanced, specific micro-history and make it relevant by incorporating a larger important historical event.”²⁷

The students accomplished this not by creating a narrative, as they would if they were writing a research paper, but by developing, and then programming, a specific rule set. By focusing on games as systems with rule sets, students can develop what this student leader called “a module to process preexisting historical data,” according to the choices of the player. The resulting gameplay experience provides new opportunities for students to express their own representations of history, in all of its complexity. As these students knew, distant geopolitical forces could turn what appeared to be a good decision into a bad one; in this case, a tavern keeper’s (player’s) attempt to diversify his business by investing in nearby farmland could result in bankruptcy if the British farmers (no longer able to sell their produce to their American customers) defaulted on their loans. These students had programmed caution into the computer code, and in this way rendered their judgment on the reasons behind the economic havoc of the war period. By expressing these historical events in computer language that was appropriate to the content, the students were able not only to capture these events, but also to provide opportunities to imagine how this story, and many others like it, might have turned out differently.

Assessment

The developers of *Brigade* and *Tavern Keeper* responded to the challenge of creating a history game with enthusiasm. For young people who have grown up using digital technologies to create content and connect with one another, the opportunity to develop a game with a group of their colleagues is strikingly refreshing. But the excitement of the project’s initial phase was quickly replaced by anxiety. Students recognized that this assignment did not fit the conventions of a typical undergraduate humanities course, in which they separately write texts addressed to their professors. How could a team of students develop a game to be played by their peers?

These kinds of concerns are shared by students learning through games, whether they are playing, modding, or building ex novo; how, they ask, will this project be marked? The students in our courses have three main concerns: (1) How will they receive regular feedback on their progress? (2) How will they know what is being marked? (3) In the case of group projects, how will the instructors ensure that students’ grades reflect their participation in the group project? We shared our students’ first concern about feedback, but from a different perspective: how will we manage, and mark, the mass of material that is produced as a by-product? In the end, we have developed a marking scheme that ensured that the students received regular responses

to their game development. In Kevin Kee's third-year course, for instance, where the game counts for 30 percent of the student's final grade in the course, each step of development is graded: the Draft Proposal, due at the end of November, out of 5; the Design Document, due at the end of February, out of 5; the report from the first round of testing, at the beginning of March, out of 5; the report from the second round of testing, near the end of March, out of 5; the final product, due in April, out of 10. In this way, students are given opportunities to work with the professors and teaching assistants to correct errors as they emerge.

Our students are also concerned with how their performance is being measured. In a first-year course taught by Shawn Graham, in which students played a scenario that had been built in *Civilization IV*, the assessment structure included a "game diary" that asked specific questions of the students at particular points in the game, forcing them to reflect on the anachronisms and other artifacts of the gameplay that corresponded or conflicted with their previous readings of history. The game diary was intended to replace one of the assigned essays, but in the end, every student opted to hand in a traditional essay. The reluctance to "play for marks" was partly one of academic culture, but also one of explication: the students did not know what was to be marked when they played the scenario.

The solution we propose is familiar to many teachers of technical courses: rubrics. These have the advantage of distilling the marking to a checklist of criteria. Students can see at a glance what the professor expects them to do to achieve a superior grade (see table 13.3).

For a project in which a student is developing a scenario in *Civilization IV*, the rubric defines expectations according to choice of subject, research, preparation for development, appropriateness of the subject to the medium, collaboration, recognition of the limits of the medium, and facility with the technology. The first criterion addresses the question: has the student selected a good problem to try to render in a scenario? (As noted above, *Civilization IV* has built-in assumptions about how history unfolds. Does the proposed scenario play to those assumptions, or does it challenge them?) The second criterion assesses whether the student has assembled the appropriate primary and secondary sources to ensure the authenticity of the scenario that she intends to develop (and a very good student will explore what makes for an authentic scenario). The next two criteria are aimed at the student's preparation: Has the student addressed the key issues inherent to the content to be modded? Has she chosen, for instance, an appropriate place to map, with an appropriate geographical and time scale? The following criteria assesses whether the student recognizes the problems inherent to developing a historical mod. The "uses forum/wiki" criterion highlights the

TABLE 13.3. Rubric for assessing the development of a scenario in *Civilization*

Criterion	Level 4	Level 3	Level 2	Level 1
Analyze Given Scenario	Analysis of given scenario provides thorough insight	Provides considerable insight	Some insight	Limited or superficial insight
Assess Research Information	Assessment of research information demonstrates thorough use of criteria	Considerable	Some	Limited
Communicate Design Ideas	Communicates these with a high degree of clarity	Considerable	Some	Limited
Create a Design Plan	Creates a design plan with excellent organization	Considerable	Some	Limited
Demonstrate Understanding of the Topic	Demonstrates a thorough understanding	Considerable	Some	Limited
Use Forum/Wiki to Communicate Information to Members of the Group, Teacher	Uses these tools with excellent success	Considerable	Some	Limited
Identify Design Issues	Thoroughly identifies design issues	Substantially identifies design issues	Adequately identifies design issues	Briefly identifies design issues
Manipulate Game Engine	Is able to use (or write) more complex programs or the worldbuilder to manipulate scenario with facility	With good success	With some success	With limited success
Use Appropriate Software to Document a Complete Project	Chooses appropriate software and uses special features to thoroughly document all parts of the project	Well documented	Is able to use a word processor to complete most of the documentation	Is able to use a word processor to complete parts of the documentation

expectation that students will collaborate in the forum or wiki developed for them, and offer help to one another as they design their scenarios. The “identify design issues” criterion forces the student to demonstrate that she is aware of the constraints of the *Civilization IV* environment. And the last two criteria focus on the tool. Just as a student who is incapable of writing will be judged for his grammar and syntax, a student modding a game will be evaluated for the manner in which he uses the computer language that he, in this course, has been trained in.

The final criteria highlight the challenge for students who are not literate with the software, and the computer languages that make them operate. How do we incorporate traditional humanities students into these projects? From the other side of the academic spectrum, how do we include in these projects a student who has a strong grasp of C++, xml, or scenario building, but is weak in history? In Kee’s course, the solution was to assemble teams of students who together possessed the requisite knowledge and skills to develop a game. But this solution raised another problem: how, to highlight the students’ third concern, would the development of their project be assessed in a manner that was fair to each student, some of whom might take on a high degree of responsibility, and others of whom might decide to let their peers do all of the work? The solution, in this case, was for the instructor to acknowledge that the students were best fit to determine the participation of their peers, and thus to award a mark to the assignment—for example, 25/30, and then let the students divide that mark according to their assessment of one another. If there were 10 students in a group, they would be apportioned a total of 250 marks for this stage (10 students $\times$ 25 marks), which they could then award to each of the members of their team. A student who decided to opt out of the last two months of development might be relegated to a mark of 15/25, which would allow for a student who had devoted 40 hours a week to the project to be awarded 35/25 (an unlikely, but possible outcome). Each student’s total mark was then averaged (divided by 10) to determine the final mark (out of 25). The distribution of marks occurred anonymously, and anomalies (for example, involving two students who disliked each other, and awarded each other zeros) were easy to recognize, and if necessary, address.

Conclusion

New approaches to teaching history in an age of pervasive computing will necessitate inventive forms of assessment. And an innovative frame of mind will be needed as technology becomes more pervasive, and the manner in which we teach history changes further. For instance, the constant creep of computing into our daily lives is presently in the process of liberating gaming

from the confines of computer monitors and tabletops, as augmented reality games bring together game-worlds and everyday life.

For instance, Kevin Kee's *Niagara 1812: Return of the Fenian Shadow* and *Queenston 1812: The Bomber's Plot* depend on smart-phone enabled GPS to provide clues to guide the player, via iPhone, around the historic core of the villages of Niagara-on-the-Lake and Queenston. In these environments, fictitious events occurring in the virtual world intersect with real-world space to create the game.

The challenge for teachers is that these games require students to be on-site in Niagara-on-the-Lake or Queenston, outside the safe boundaries of the school proper. More accessible to students in school are a new genre of games that treat the web like a physical space. *The Nethernet* (formerly *PMOG*) is a game that is played by clicking from website to website, following the prompts of a guide. While *The Nethernet* itself uses the metaphor of a "mission" to describe what is going on in the game (the idea being that other players will try to disrupt or enhance your movement through the Internet by laying mines that "blow up" your browser window, or portals that take you to unrelated websites), the players themselves often use the metaphor of a "tour" (with disruption kept to a minimum). Often, missions are really guided tours of specialty websites.²⁸

Unlike standard games, the environments that *The Nethernet* or *Niagara 1812* are played in can themselves change the experience of the game. Websites are taken down, links become broken, it may rain in Niagara-on-the-Lake, or a street may be closed for repair. The game environments are outside the control of the game maker. Even so-called persistent world games like *World of Warcraft* have underlying structures controlled by the game maker. This means that for augmented reality games, the experience of the player, and the player's response, can never be fully accounted for and so "gaming the game" is part of the main gameplay: the meta-game and the game intersect. Finally, there is never a point where you have won a game of *The Nethernet*. You may complete the mission, the tour, but you are rewarded with points toward gaining another experience level (which are in practice infinite). In *Niagara 1812* the game ends once and for all when you have finished the story.

The act of playing these games in this manner causes the player to engage with the material in a way she had not before: she is looking at a series of related sites from the perspective of the creator of the mission. One could imagine a student of Roman archaeology creating a *Nethernet* mission on curse-tablets. The mission might begin as a simple show and tell. Other students could then play the mission, leaving mines on pages they think are "bad" (poor information, bad research, whatever) or portals to "good" sites: the environment of the gameplay changes as the players play (like the rink in a hockey game gets chewed up by the skaters, influencing the way the puck

bounces or skips across the ice). Inserting puzzles into the mission would force a deeper engagement still, and completing a puzzle mission would constitute a formative assessment exercise. Afterward, since the game records the play, another part of the meta-game would come in classroom discussions.

In this way, students play the meta-game. Educators have long been concerned that students do not adequately analyze information they find on a website. *The Nethernet* asks the student to treat web pages not as tools, but as artifacts to be analyzed. We began this chapter by contending that students may benefit from analyzing games, in the same way that they decode texts, and that students must recognize implicit viewpoints in games, just as they do in essays and books. And we showed that students can learn to build their own theses, as they do in the writing of historiographical and research papers, through the act of modding and building a game. Given that the creation of expressive content is among the primary forms of digital engagement by our students, it is an appropriate way to engage technology in an age of pervasive computing.

NOTES

1. "Pervasive computing" can also refer to the distribution of computing power across multiple platforms or devices (as in the *SETI@home* project, which divides up enormous computing problems into micro-chunks, and uses unused computing cycles in idle computers to perform calculations). We use the term to capture both ideas: that "pervasive computing" is distributed computing power across multiple devices and platforms, and into every corner of our lives.

2. *Open Knowledge and the Public Interest* (University of California at Berkeley, n.d.), accessed July 31, 2012, <http://okapi.wordpress.com/2008>.

3. Hannah Green and Celia Hannon, *Their Space: Education for a Digital Generation* (London: Demos, 2007), 9–10.

4. "List of Best-Selling Video Games," accessed November 15, 2012, http://en.wikipedia.org/wiki/List_of_best-selling_video_games#PC.

5. Keith Jenkins and Alun Munslow, *Re-thinking History* (London: Routledge, 2003).

6. David Lowenthal, *The Heritage Crusade and the Spoils of History* (Cambridge: Cambridge University Press, 1998).

7. Edward Castronova, *Synthetic Worlds: The Business and Culture of Online Games* (Chicago: University of Chicago Press, 2005), 3, 291–92.

8. Christian Elverdam and Espen Aarseth, "Game Classification and Game Design," *Games and Culture* 2, no. 1 (January 1, 2007): 3–4.

9. Tom Taylor, "Historical Simulations and the Future of the Historical Narrative," *Journal of the Association for History and Computing* 6, no. 2 (2003).

10. Kurt Squire, "Replaying History: Learning World History through Playing Civilization III" (Ph.D. dissertation, Indiana University, 2004).

11. Kevin Schut, "Strategic Simulations and Our Past," *Games and Culture* 2, no. 3 (July 1, 2007): 213–35; William Uricchio, "Simulation, History, and Computer

Games,” in *Handbook of Computer Game Studies*, ed. Joost Raessens and Jeffrey Goldstein (Cambridge, Mass.: MIT Press, 2005), 327–38; Esther McCallum-Stewart and Justin Parsler, “Controversies: Historicising the Computer Game,” in *Situated Play: Proceedings of DiGRA 2007 Conference* (Tokyo: Digital Games Research Association, 2007), 203–10.

12. Martin Ryle, “Humanist mailing list 2: 592,” n.d.; David Kushner, “In Historical Games, Truth Gives Way to Entertainment,” *The New York Times*, September 6, 2001, sec. Technology, accessed July 31, 2012, <http://www.nytimes.com/2001/09/06/technology/in-historical-games-truth-gives-way-to-entertainment.html>; Kacper Poblocki, “Becoming-State: The bio-Cultural Imperialism of Sid Meier’s Civilization,” *Focaal—European Journal of Anthropology* 2009, no. 39 (2002): 163–77; Uricchio.

13. Robert MacDougall, “Madness and Civilization III,” *Old is the New New*, n.d., accessed July 31, 2012, <http://www.robmacdougall.org/index.php/2007/07/madness-and-civilization-iii/>.

14. Ian Bogost, *Persuasive Games: The Expressive Power of Videogames* (Cambridge, Mass.: MIT Press, 2007), 28–44.

15. Uricchio, 336.

16. Jesper Juul, “Without a Goal,” in *Videogame, Player, Text*, ed. Barry Atkins and Tanya Krzywinska (Manchester: Manchester University Press, 2008), accessed July 31, 2012, <http://www.jesperjuul.net/text/withoutagoal/>.

17. Juul; Mihaly Csikszentmihalyi, *Flow: The Psychology of Optimal Experience* (New York: HarperPerennial, 1991), 49–50.

18. In a previous article (Kevin Kee, “Computerized History Games: Options for Narratives,” *Simulation & Gaming* 42, no. 4 (August 2011), Kee contended that history can be represented in different ways using this typology: with historical epistemologies that focus on history as the mastery of a singular narrative matching well with goal-oriented games, and historical epistemologies that focus on history as the act of discovery about the past matching well with goalless games. In this chapter we go further, focusing beyond the goals within games to how we learn not just within games, but with and beyond games, in an age of pervasive computing.

19. Espen Aarseth, Solveig M. Smedstad, and Lise Sunnana, “A Multi-Dimensional Typology of Games,” *Proceedings of the Level Up Games Conference* (Utrecht, Netherlands Digital Games Research Association, 2003), 48–53.

20. “Meta-gaming,” accessed August 1, 2012, <http://en.wikipedia.org/wiki/Metagaming>.

21. The poll was conducted by a user of the *CivFanatics* forums who went by the name of “wooter.”

22. A notable example of blog-based peer review comes from Noah Wardrip-Fruin, accessed July 31, 2012, <http://grandtextauto.org/2008/04/03/blog-based-peer-review-some-preliminary-conclusions-part-1/>.

23. *CivFanatics*, accessed July 31, 2012, <http://forums.civfanatics.com/showthread.php?t=187436>.

24. Idit Harel and Seymour Papert, *Constructionism* (New York: Ablex, 1991).

25. The project website has been taken down, though the Interactive Arts and Science program’s page was available as of July 31, 2012, <http://www.brocku.ca/humanities/departments-and-centres/interactive-arts-and-science>.

26. Interactive Arts and Science, accessed August 1, 2012, <http://www.brocku.ca/humanities/departments-and-centres/interactive-arts-and-science>.

27. Tom Mitrovic, reflection paper (unpublished student essay, Brock University, n.d.).

28. Alas, at the time of publication, *The Nethernet* was no longer online.